UNIVERSITY OF RUHUNA

BACHELOR OF INFORMATION AND COMMUNICATION TECHNOLOGY DEGREE

LEVEL II (Semester I)

ICT2132- Object Oriented Programming Practicum

Mini Project- Software Requirements Specification [SRS]

Submitted By: Group OOPify

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Software Requirements Specification (SRS)

Faculty of Technology Management System

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1. Introduction

1.1 Purpose

This document describes the requirements for a Java and MySQL system to manage faculty operations including user profiles, courses, marks, attendance, notices, timetables, and medical records.

1.2 Scope

The system will be used by Admin, Lecturers, Technical Officers, and Undergraduates at the Faculty of Technology.

The system will manage:

- User Profiles (Admin, Lecturer, Technical Officer, Undergraduate)
- Course Details
- Attendance Management
- Marks & GPA Calculation
- Eligibility Checking
- Medical Records
- Timetables
- Notices

The system will help automate academic record management and improve efficiency in handling undergraduate data.

1.3 Definitions

Term	Meaning
CA	Continuous Assessment
GPA	Grade Point Average
SGPA	Semester GPA
CGPA	Cumulative GPA
Eligibility	Qualification to sit final exam (Attendance ≥ 80% and CA ≥ 40%)

2. User Roles and Permissions

2.1 Admin

- Create and manage all user profiles (Admin, Lecturers, Technical Officers, Undergraduates)
- Create and manage Courses
- Create and manage Notices
- Create and manage Timetables

2.2 Lecturer

- Update own profile (except username and password)
- Add and modify course materials
- Upload marks for all exams
- View undergraduate details
- Check undergraduate eligibility (CA >= 40%)

- View undergraduate marks, grades, and GPA
- View attendance and medical records
- View notices

2.3 Technical Officer

- Update own profile (except username and password)
- Add and manage attendance details
- Add and manage medical details
- View notices
- View department timetables

2.4 Undergraduate

- Update contact details and profile picture only
- View own attendance
- View own medical details
- View own course details
- View own grades and GPA
- View own timetables
- View notices

3. Functional Requirements

3.1 User Management

Admin shall:

- Create, update and delete user profiles
- Store:

- 01 Admin
- At least 05 Lecturers
- At least 04 Technical Officers
- At least 20 Undergraduates

3.2 Course Management

Admin shall:

- Create and maintain courses
- Assign lecturers to courses
- Store theory and practical components

3.3 Attendance Management

Technical Officer shall:

- Record attendance for 15 theory sessions
- Record attendance for 15 practical sessions
- Record medical submissions
- Mark medicals as approved/not approved

System shall:

- Calculate attendance percentage
- Show:
    - 80%
    - Exactly 80%
    - < 80% without medical
    - 80% with medical
    - < 80% with medical

- Show summary (batch-wise and individual)
- Show:
    - Theory only
    - Practical only
    - Combined attendance

3.4 Marks Management

Lecturer shall:

- Enter CA marks (out of 100)
- Enter final marks (out of 100)

System shall:

- Check CA eligibility (≥ 40%)
- Show batch and individual summaries
- Calculate final marks
- Generate grades
- Calculate SGPA and CGPA

3.5 Eligibility Checking

System shall:

- Check eligibility based on:
    - Attendance ≥ 80%
    - CA ≥ 40%
- Display eligibility for:
    - Individual
    - Whole batch

3.6 Grade Management

System shall:

- Assign grades according to UGC Commission Circular No. 12-2024
- Show grades:
    - Per subject
    - Individual
    - Whole batch

3.7 GPA Calculation

System shall:

- Calculate SGPA
- Calculate CGPA
- Display subjects, grades and GPA for:
    - Individual
    - Whole batch

3.8 Notices and Timetables

Admin shall:

- Create and update notices
- Create and update timetables

All users shall:

- View relevant notices
- View relevant timetables

4. Technical Requirements

4.1 Programming Requirements

- Language: Java
- Database: MySQL
- Database connection: JDBC

4.2 OOP Concepts to Demonstrate

Each member must show use of:

- Classes and Objects
- Inheritance
- Abstraction
- Polymorphism
- Encapsulation
- Error/Exception Handling
- Database Handling
- GUI components

4.3 Non-Functional Requirements

- User-friendly interface
- Role-based access control
- Input validation
- Error messages for invalid actions
- Data persistence in MySQL

5. Deliverables and Deadlines

Task	Due Date
SRS Document	01st March 2026
Class Diagram + ER Diagram	08th March 2026
Complete Project + Report	26th April 2026
Presentation	To be announced

6. Conclusion

The Faculty Management System will provide a structured and efficient way to manage academic records of the Faculty of Technology at the University of Ruhuna.

The system ensures:

- Accurate attendance tracking
- Proper eligibility checking
- Transparent grading and GPA calculation
- Efficient user management

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